

A jewelry store owner tracked the number of pieces she sold for 22 days. Her data are shown.

8	3	9	5	6	4	9	15	5	7	7
4	5	8	12	8	6	6	5	10	8	11

1. Create a frequency table of the data.

Number of pieces sold	Frequency
3 – 6	7
6 – 9	9
9 – 12	4
12 – 15	1
15 – 18	1

2. Determine the five-number summary of the data set.

Min: 3

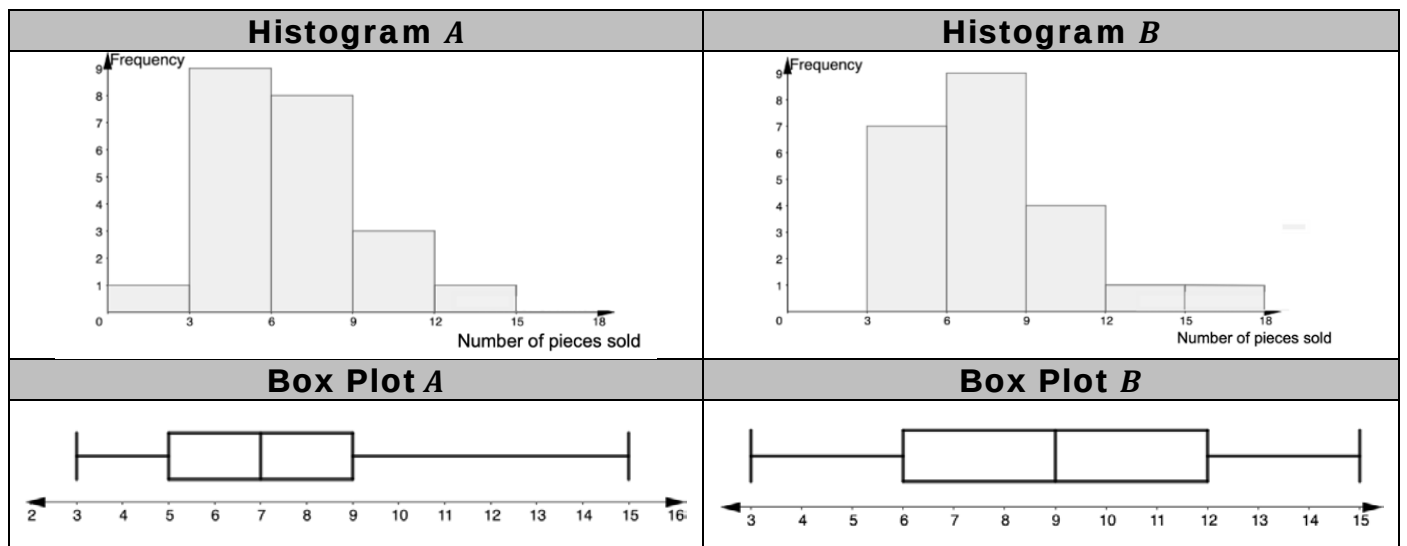
Q₁: 5

Median: 7

Q₃: 9

Max: 15

Two histograms and two box plots are shown.



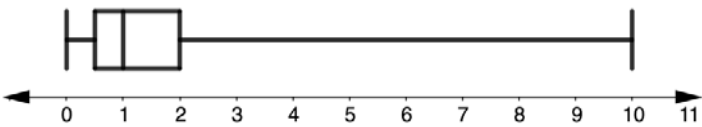
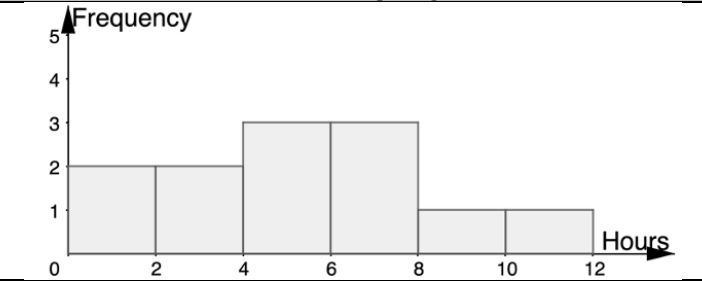
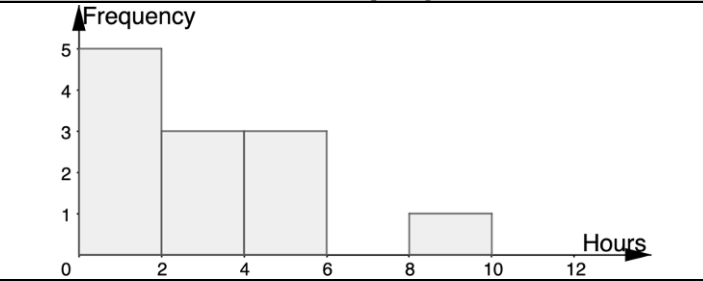
3. Explain which histogram better represents the data.
Histogram B matches the frequency table with the same intervals and frequencies.
4. Explain which box plot better represents the data.
Box plot A has the correct median and quartiles, unlike B.

5. Which graph, box plot or histogram, is better for displaying this data? Explain your reasoning.

Explanations may vary.

The box plot would give the store owner a better idea of the average amount of pieces she could expect to sell.

Four data displays and four data sets are shown.

Data Display 1	Data Display 2
	
Data Display 3	Data Display 4
	
Data Set A	Data Set B
0, 0, 2, 3, 4, 4, 4, 6, 6, 7, 8, 10	0, 1, 2, 3, 4, 5, 5, 6, 7, 8, 9, 10
Data Set C	Data Set D
0, 0, 1, 1, 1, 2, 2, 3, 4, 5, 5, 9	0, 0, 0, 1, 1, 1, 1, 1, 2, 2, 5, 10

Match each Data Display to a Data Set.

6. Display 1: D 7. Display 2: B

8. Display 3: A 9. Display 4: C

10. Explain how you matched the data set that was represented by Data Display.

Explanations may vary.

First, I found the medians for each set of data which matched the data to the box plots (and then double-checked quartiles). Then the last two data sets, I checked the frequency using the intervals from the histogram and matched them.